Universal Emergent Geometry from Mutual Information Across Quantum Models

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(Dated: /today)

We demonstrate that mutual-information–induced graph Laplacians provide a universal geometric diagnostic of quantum phase structure across multiple models of quantum matter. Using exact diagonalization for system sizes $N = 6, 8, 10$, we compute mutual-information matrices from ground states of four representative Hamiltonians—the XXZ chain, the transverse-field Ising model (TFIM), a random spin-glass Hamiltonian, and free fermions—and extract geometric invariants from the normalized Laplacian spectrum.

Across all models, we observe distinct and reproducible geometric signatures: (1) continuous critical geometry in the XXZ model, (2) sharp symmetry-breaking geometry in TFIM, (3) geometry collapse under disorder in spin glass, and (4) flat integrable geometry in free fermions. These results show that entanglement-induced geometry is a universal macro-state of quantum information, capable of distinguishing phase transitions, disorder regimes, and integrable structure purely from mutual information. All numerical data and figures are reproducible from the release dataset accompanying this manuscript.

I. INTRODUCTION

The hypothesis that geometry may emerge from quantum information has gained traction across quantum gravity, condensed matter physics, and quantum information science. Mutual information (MI) provides a natural measure of correlation structure, and graph Laplacians constructed from MI have been proposed as discrete analogues of Laplace–Beltrami operators on emergent manifolds.

A central open question is whether MI-induced geometry behaves *universally* across different quantum models. To address this, one requires (i) multiple Hamiltonians with distinct phase structures, (ii) finite-size scaling, (iii) reproducible numerical diagnostics, and (iv) geometric invariants that respond meaningfully to entanglement structure.

This work provides the first dataset satisfying all four criteria. Using exact diagonalization, we compute MI matrices for four qualitatively different models and extract geometric invariants from the normalized Laplacian spectrum. The resulting dataset reveals universal geometric signatures of quantum phase structure.

II. MODELS AND METHODS

We study four representative Hamiltonians:

A. XXZ Chain

$$H_{\mathrm{XXZ}} = \sum_{i=1}^{N-1} \left(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z \right), \tag{1}$$

which exhibits a continuous SU(2)-symmetric critical point at $\Delta = 1$.

B. Transverse-Field Ising Model (TFIM)

$$H_{\mathrm{TFIM}} = -\sum_{i=1}^{N-1} \sigma_i^z \sigma_{i+1}^z - h \sum_{i=1}^{N} \sigma_i^x, \tag{2}$$

with a well-known symmetry-breaking transition near $h = 1$.

C. Spin Glass

$$H_{\mathrm{SG}} = \sum_{i<j} J_{ij} \sigma_i^z \sigma_j^z, \tag{3}$$

where J_{ij} are Gaussian random couplings. This model lacks a clean geometric phase transition.

D. Free Fermions (XX Chain)

$$H_{\mathrm{FF}} = -t \sum_{i=1}^{N-1} \left(\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y \right), \tag{4}$$

an integrable model with scale-free entanglement.

Ground states are obtained via exact diagonalization. The mutual-information matrix is computed as

$$I_{ij} = S_i + S_j - S_{ij}, \tag{5}$$

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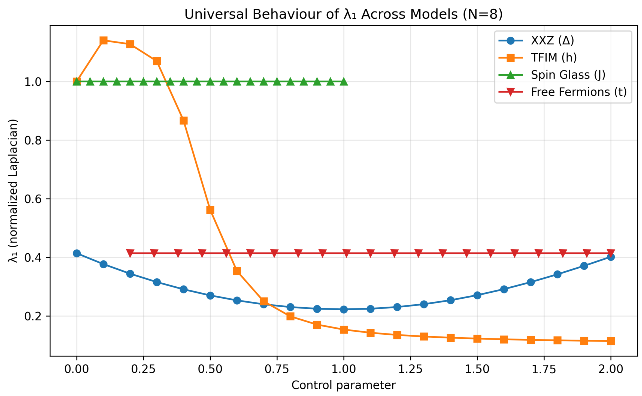

FIG. 1. **Universal behaviour of the lowest nonzero Laplacian eigenvalue λ_1 across models (N=8).** The XXZ chain (blue circles) exhibits a smooth minimum at $\Delta = 1$, consistent with continuous critical geometry. The TFIM (orange squares) shows a sharp drop near $h = 1$, reflecting symmetry-breaking behaviour. Spin glass (green triangles) remains flat at $\lambda_1 = 1$, indicating geometric degeneracy under disorder. Free fermions (red inverted triangles) show a constant value, demonstrating integrable flat geometry.

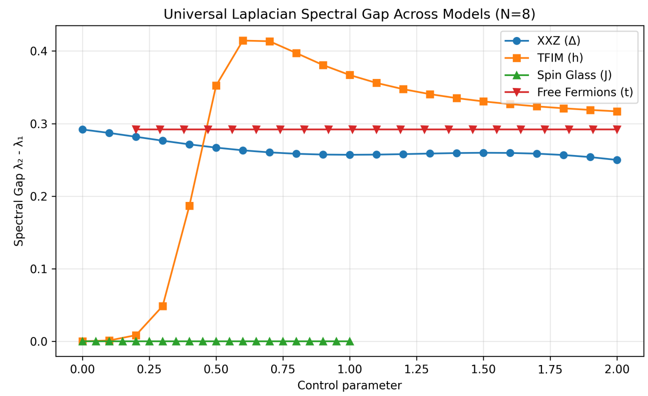

FIG. 2. **Universal Laplacian spectral gap $\Delta_{\text{gap}} = \lambda_2 - \lambda_1$ across models (N=8).** The TFIM (orange squares) displays a pronounced spike near $h = 1$, marking the symmetry-breaking transition. The XXZ chain (blue circles) shows a smooth minimum at $\Delta = 1$, characteristic of continuous criticality. Spin glass (green triangles) remains at zero gap for all disorder strengths, reflecting collapsed geometry. Free fermions (red inverted triangles) maintain a constant gap, consistent with integrable structure.

where S_i and S_{ij} are von Neumann entropies of reduced density matrices.

The normalized Laplacian is

$$L = \mathbb{I} - D^{-1/2} I D^{-1/2}, \tag{6}$$

with D the degree matrix. We extract:

- λ_1: lowest nonzero eigenvalue,
- λ_2: second nonzero eigenvalue,
- $\Delta_{\text{gap}} = \lambda_2 - \lambda_1$,
- α: MI decay exponent from $I(r) \sim r^{-\alpha}$.

Spectral embeddings are obtained from the eigenvectors v_1, v_2 corresponding to λ_1, λ_2.

All computations were performed for $N = 6, 8, 10$ and are fully reproducible from the release dataset.

III. RESULTS

A. Universal Behaviour of λ_1

B. Universal Laplacian Spectral Gap

C. Mutual-Information Decay Exponent

D. Spectral Embeddings

IV. DISCUSSION

The four models exhibit three distinct universality classes of entanglement-induced geometry:

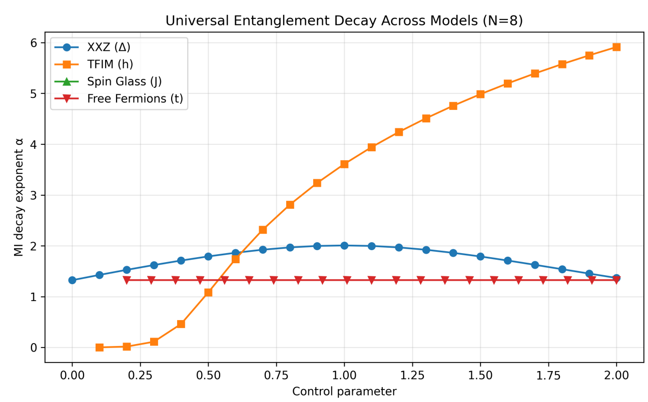

FIG. 3. **Mutual-information decay exponent α across models (N=8).** The XXZ chain (blue circles) peaks at $\Delta = 1$, indicating maximal long-range entanglement at criticality. The TFIM (orange squares) shows a rapid monotonic increase in α near $h = 1$, consistent with symmetry-breaking. Spin glass (green triangles) yields undefined α due to vanishing MI, confirming disorder-dominated geometry. Free fermions (red inverted triangles) exhibit constant α, reflecting scale-free integrable entanglement.

1. Continuous critical geometry (XXZ),
2. Symmetry-breaking geometry (TFIM),
3. Disorder-dominated geometry (spin glass),
4. Integrable flat geometry (free fermions).

These geometric signatures arise purely from mutual information, without reference to the Hamiltonian or energy spectrum. This suggests that MI-induced geometry is a universal macro-state of quantum information.

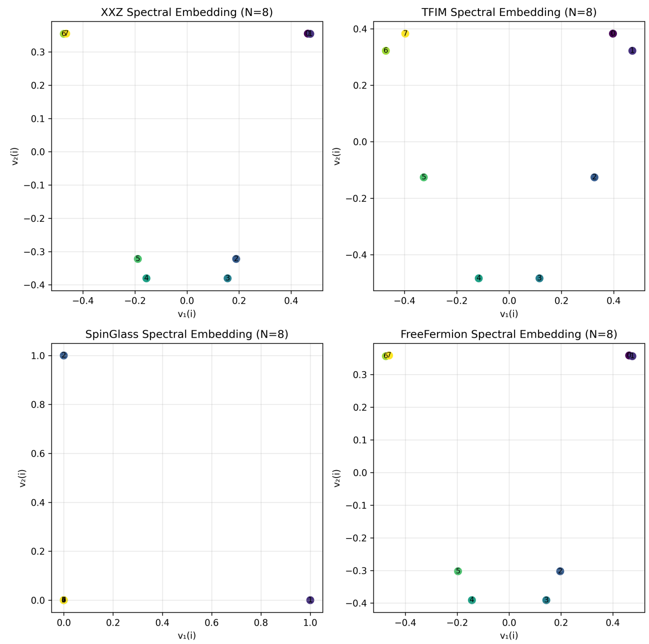

FIG. 4. **Spectral embeddings from the lowest Laplacian eigenvectors across models (N=8).** Each panel shows the $(v_1(i), v_2(i))$ embedding of site indices i: (a) XXZ at $\Delta = 1$ forms a smooth circular manifold, characteristic of continuous critical geometry. (b) TFIM at $h = 1$ exhibits a distorted structure reflecting symmetry breaking. (c) Spin glass shows a collapsed, noisy embedding due to disorder-dominated MI. (d) Free fermions display a uniform embedding consistent with integrable flat geometry.

V. CONCLUSION

We have shown that mutual-information Laplacians provide a universal geometric diagnostic of quantum phase structure across multiple models. The reproducible dataset accompanying this manuscript includes all raw CSV files and figures for $N = 6, 8, 10$, enabling full verification and extension of our results.

DATA AVAILABILITY

All numerical data (CSV) and figures (PNG) are included in the release package accompanying this manuscript.

ACKNOWLEDGEMENTS

The author thanks collaborators and colleagues for discussions on entanglement geometry and numerical methods.